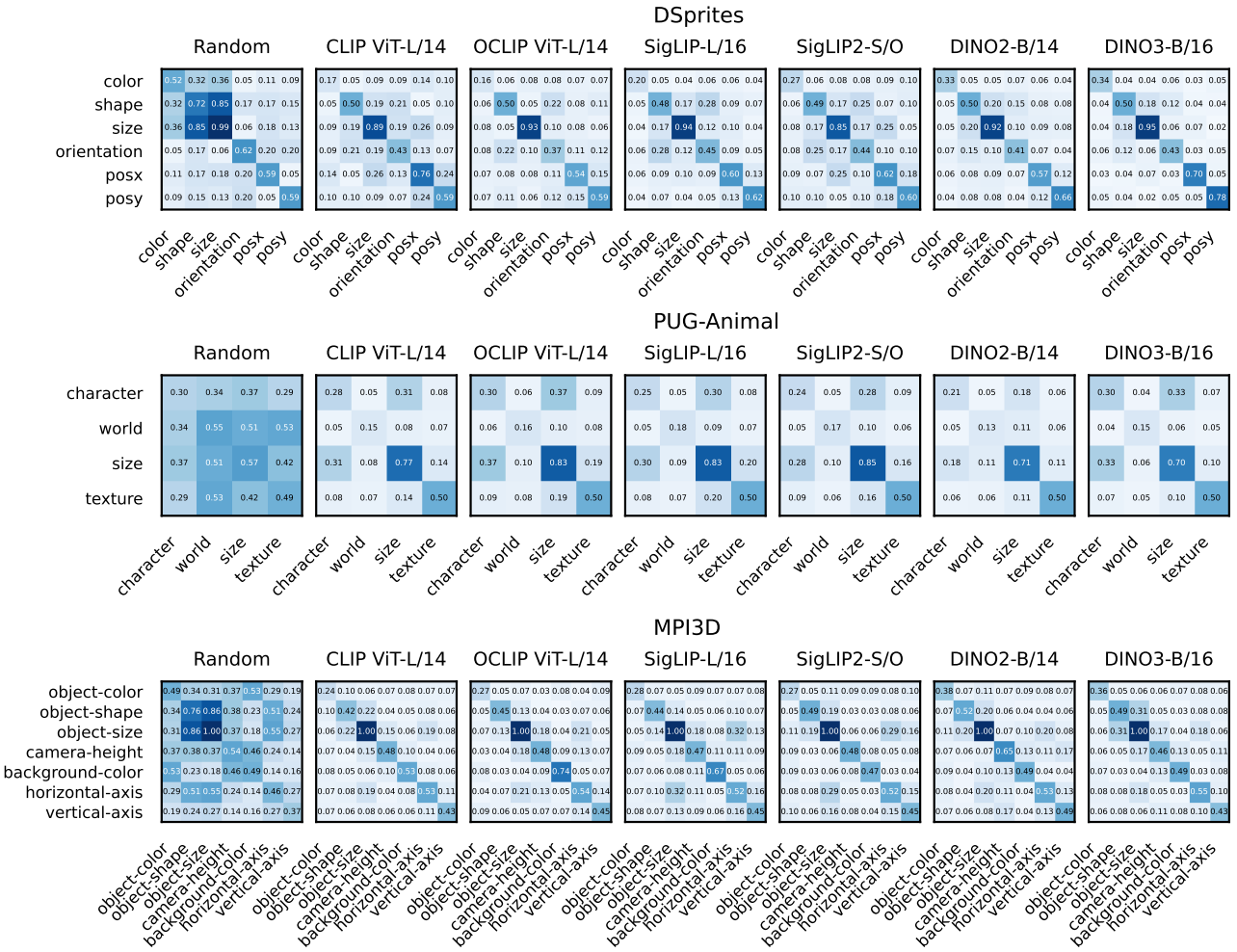


Figure 24. **Full orthogonality matrices across models and datasets.** Each entry reports mean absolute cosine between centered factor directions. Diagonal blocks correspond to within-concept similarity; off-diagonal blocks correspond to across-concept similarity. Lower off-diagonal values indicate stronger cross-concept orthogonality.

We summarize the same trend in aggregate form in Fig. 25, comparing within-concept and across-concept values directly.

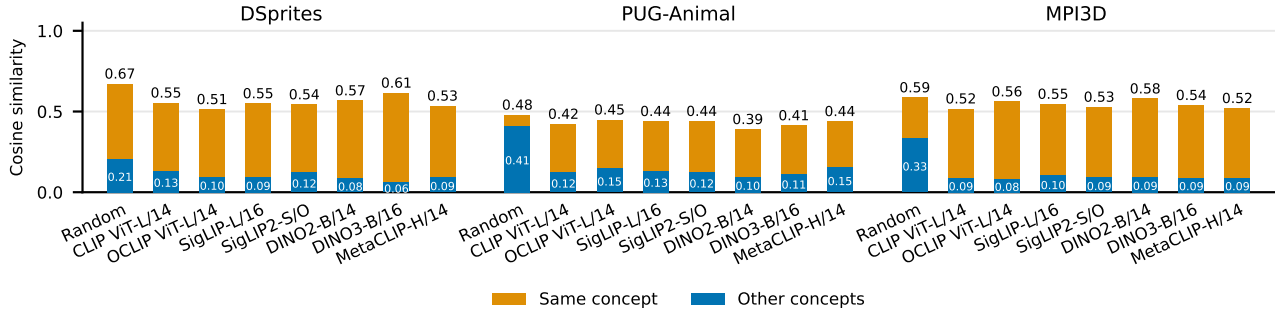


Figure 25. **Aggregated orthogonality summary.** Bars compare within-concept direction similarity and across-concept similarity; across-concept values are consistently lower, matching the orthogonality pattern predicted by the theory.

I.3. Dimensionality of factors

This subsection provides the full model dimensionality results that complement Section 5.4. For each model and dataset, we estimate per-concept effective dimensionality as the minimum number of PCA components needed to explain 95% variance in recovered factors, and report it relative to concept cardinality. The complete per-model plots are shown in Fig. 26.

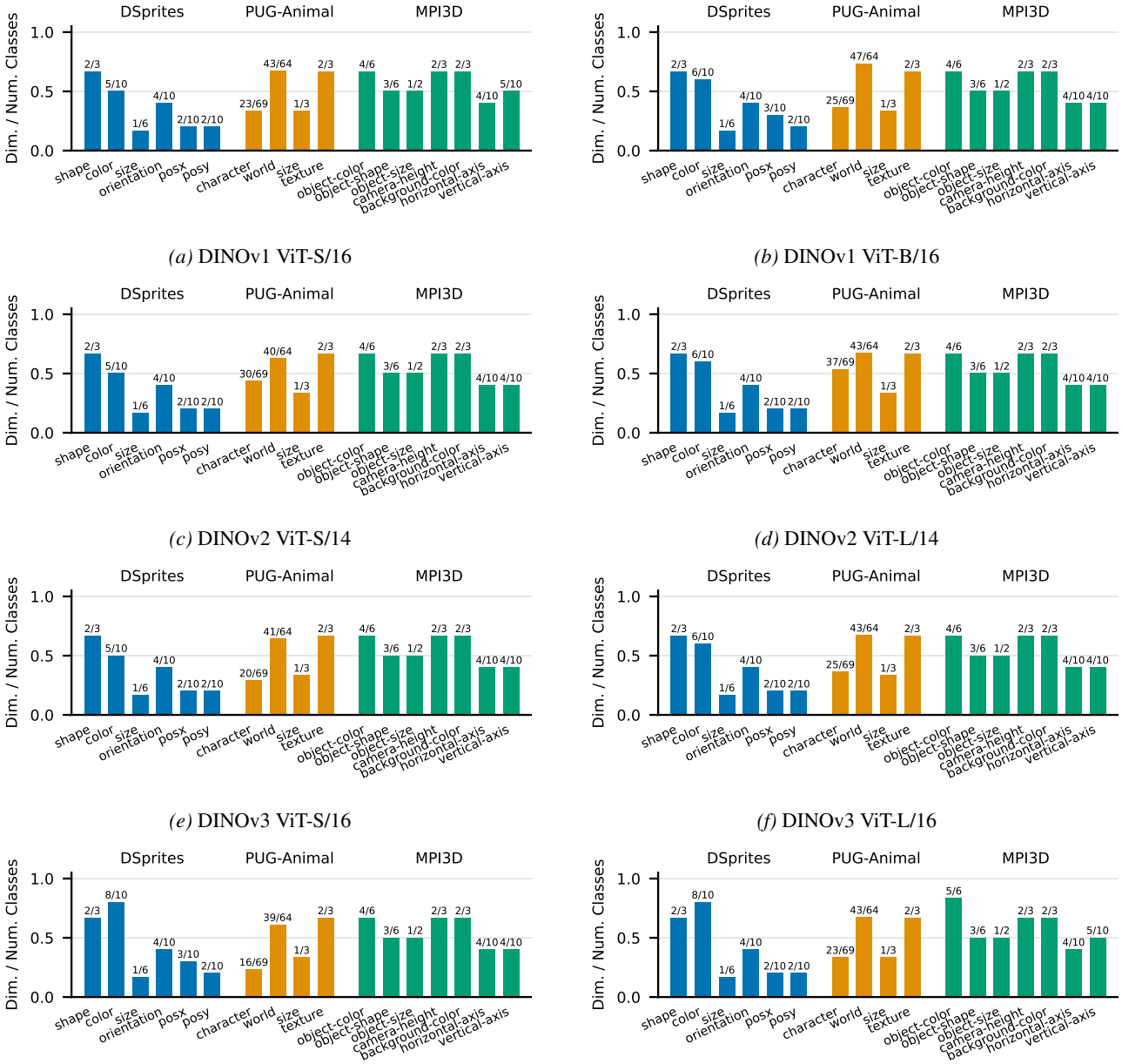
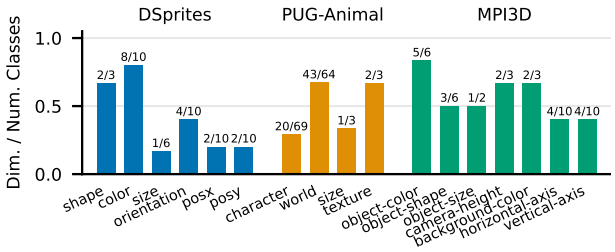
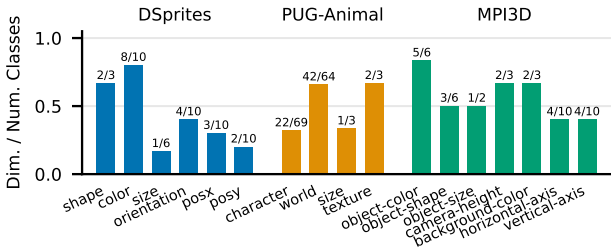


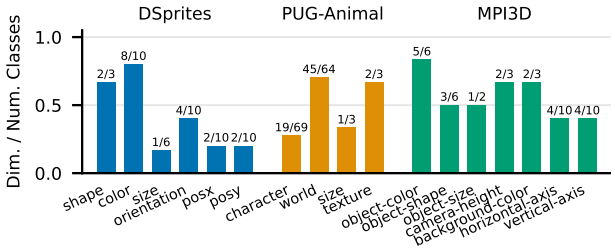
Figure 26. Dimensionality results computed as the number of SVD factors required to reach 95% explained variance, per dataset, across representative vision(-language) backbones.



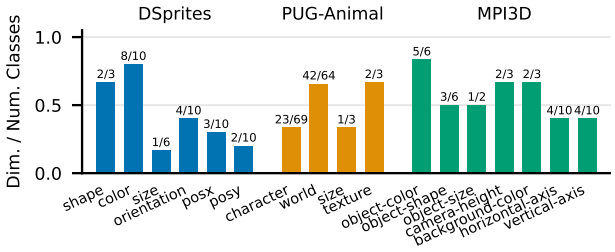
(i) OpenCLIP ViT-B/16 (LAION-400M)



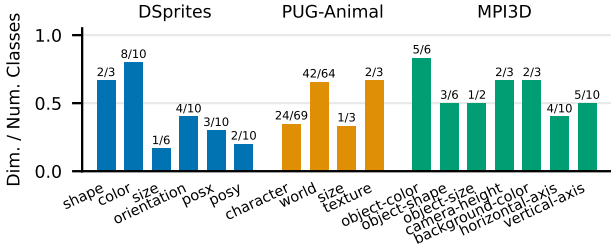
(j) OpenCLIP ViT-L/14 (LAION-2B)



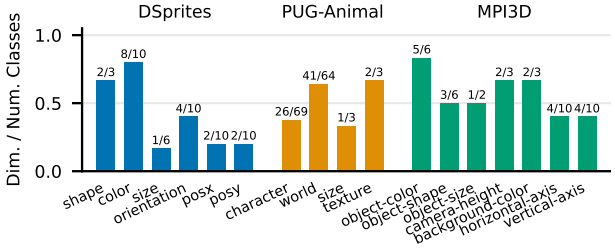
(k) MetaCLIP 2.5B ViT-B/32



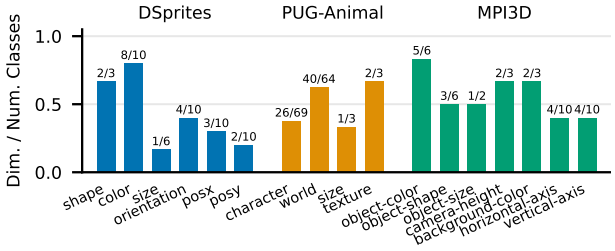
(l) MetaCLIP 2.5B ViT-L/14



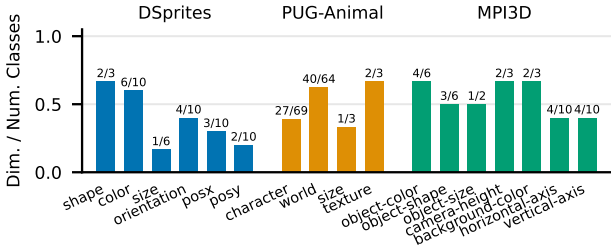
(m) MetaCLIP2 ViT-H/14 (QuickGELU)



(n) MetaCLIP2 ViT-bigG/14



(o) SigLIP-L/16



(p) SigLIP2 SO400M/16 (256)

Figure 26. Dimensionality results (continued from Fig. 26).